

HW3L

A Three-Field Hu–Washizu Tetrahedron in Logarithmic Strain

Research note

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Abstract

HW3L (Hu–Washizu, three fields, logarithmic strain) is a tetrahedral element for large-deformation deviatoric plasticity, aimed at removing the stabilization that two families of methods carry: the composite tetrahedron, a tetrahedron subdivided into four linear tetrahedra with one element-constant pressure, which relieves volumetric locking and introduces soft modes that a penalty energy removes; and variational-multiscale methods, which add a subgrid pressure proportional to the time step that reaches the plastic return map. Three fields — the motion, a volumetric strain and a pressure — are coupled through a Hu–Washizu functional, in which the two auxiliary fields are independent of the motion and tied to it by a constraint term. The deviatoric response is evaluated pointwise at quadrature, so the constitutive law is an unmodified small-strain algorithm, and both auxiliary fields are discontinuous across elements, so their elimination is element-local. The stored energy must split exactly into volumetric and isochoric (volume-preserving) parts, which five of the seven materials implemented in the finite-element code Norma do. With both auxiliary fields in one discrete space the functional reduces exactly to mean dilatation, the replacement of the volumetric strain by its element-wise projection, and what is open is the choice of pair: the discrete space for the displacement together with the discrete space for the pressure. Volumetric locking and spurious soft modes are failures of the two hypotheses of Brezzi’s theorem for that pair: the inf-sup condition, that every discrete pressure is resisted by some discrete displacement with a constant independent of the mesh, and coercivity of the deviatoric form on the discretely isochoric subspace, that no discrete displacement the pressure does not resist has vanishing deviatoric energy. The first is settled by measurement over a mesh sequence against a control pair of established stability: of six element-local pairs only the three-dimensional Crouzeix–Raviart pair — quadratic displacement enriched with face and interior bubbles, over a linear discontinuous pressure — is stable, at a constant of 0.2968; its face bubbles are global, and the pair has 2.6 times the degrees of freedom of the composite tetrahedron. The second holds in elasticity for every conforming pair, and under a plastic tangent that removes the fraction η of the shear stiffness along the flow direction ($\eta = 1$ for perfect plasticity) an identity bounds its constant below by $(1 - \eta)/2$ for every pair, so it cannot separate them. The claim of this note is that the pairs differ in whether their soft modes are isochoric, and it is supported on the linearized operator: under a plastic tangent the softest modes of a constant-pressure element carry a dilatation of 0.20 to 0.27 that does not decay under refinement, those of the Crouzeix–Raviart pair one that decays as the square of the mesh size h , and those of the Taylor–Hood pair (continuous quadratic displacement over continuous linear pressure) are dilatation almost entirely; in a confined plastic zone with a varying flow direction the Crouzeix–Raviart pair has no spurious mode among its twenty softest at the finest mesh, and the constant-pressure pairs retain between

five and fifteen. On deformed configurations the inf-sup constant decreases with the shear of the deformation and is unchanged by its volume change. A fourteen-point degree-five quadrature rule and a nodal fifteen-node basis are sufficient, and the explicit stable step is 3.3 times smaller than a linear tetrahedron’s. The smoothness of the equivalent plastic strain in a nonlinear solve is not measured here.

1 Problem statement

Tetrahedra are the only practical choice for meshing complex geometry automatically. Their accuracy is insufficient in four regimes: near-incompressible response at bulk-to-shear modulus ratios $\kappa/\mu \lesssim 10^4$, with κ the bulk modulus and μ the shear modulus; isochoric (volume-preserving) plasticity; large rotation; and weak shocks, pressure waves of small amplitude, in materials of high κ/μ . The target is one spatial formulation for all four, with the time integrator the only difference between the quasi-static and the dynamic regimes.

Two established remedies fail, and each failure is the failure of one hypothesis of the same theorem; this section identifies the hypothesis in each case.

Constant pressure relieves locking and introduces soft modes. The composite tetrahedron, a tetrahedron subdivided into four linear tetrahedra with one element-constant pressure [12, 17], and its five-field extension [5] relieve volumetric locking — the stiffening of a discretization whose displacement space contains too few discrete isochoric fields — with that pressure. Foulk III et al. [5] report that this choice — and it is the choice, not the element — produces a deleterious soft mode “common to both the quadratic and composite tetrahedral element with constant pressure formulations,” remedied there by a convex energy penalty. The mechanism is a coercivity failure on the discretely isochoric subspace: an element-constant multiplier does not resist displacement patterns whose volume change varies within the element and averages to zero, so those patterns carry only shear energy. In elasticity every projected-pressure element has such patterns, and a stable pair keeps their energy bounded away from zero; in a confined plastic zone the shear response softens along the flow direction and the pattern can approach zero energy. §8.4 shows that on the quadratic tetrahedron these patterns are not exactly of zero energy, and §5 states the question that count does not settle.

Residual-based stabilization contaminates the deviator. Variational-multiscale stabilization of linear tetrahedra [13] introduces a subgrid pressure of the form

$$p' = c_1 \tau_s (\kappa \operatorname{Div} \dot{\boldsymbol{\varphi}} - \dot{p}), \quad \tau_s = c_2 \Delta t, \quad (1)$$

with c_1 and c_2 tuned coefficients, which is added to the stress before the constitutive update is evaluated. Two consequences follow. The spatial solution depends on the time step Δt through τ_s , so there is no quasi-static limit independent of it. The coefficient c_2 scales a term that reaches the deviator, and therefore the plastic return map — the algorithm that projects a trial stress onto the yield surface — which is the origin of the oscillations in the equivalent plastic strain, the accumulated norm of the plastic strain increments. No choice of c_2 removes this, because the term enters the deviator by construction.

The common cause. Both are failures of the hypotheses of one theorem. Brezzi’s theorem for the mixed problem asks two things of the pair — the discrete space U_h for the displacement together with the discrete space Q_h for the pressure [3, §3.4, §5.2]: the inf-sup condition, that every pressure mode be resisted by some displacement, whose failure is locking on one side and spurious pressure on the other; and ellipticity of the deviatoric form on the discretely isochoric subspace $\ker \mathbf{G}_h$, that no displacement the pressure does not resist be free, whose failure is a soft mode. The first is a property of the pair alone and is settled in §8.5. The second depends on the material tangent as well as on the pair, and §5 states what this note claims about it. Neither difficulty is a stabilization problem: the remedy for the first is a pair satisfying the inf-sup condition uniformly in the mesh size, and the second must then be checked where the material can violate it.

2 Continuum formulation

This section states the kinematics, the stored energy, the three-field functional, its reduction to a displacement functional, the requirements it places on the material, and the stress it produces.

2.1 Kinematics and the volumetric–isochoric split

A body $\Omega \subset \mathbb{R}^3$ undergoes a motion $\mathbf{x} = \boldsymbol{\varphi}(\mathbf{X}, t)$ with deformation gradient $\mathbf{F} := \text{Grad } \boldsymbol{\varphi}$ and Jacobian $J := \det \mathbf{F} > 0$. The multiplicative factorization

$$\mathbf{F} = J^{1/3} \bar{\mathbf{F}}, \quad \bar{\mathbf{F}} := J^{-1/3} \mathbf{F}, \quad \det \bar{\mathbf{F}} = 1, \quad (2)$$

separates the two kinematic contributions this formulation treats differently: J carries everything the mixed treatment exists for, and $\bar{\mathbf{F}}$ carries everything the plasticity acts on.

The mixed field is a scalar function of J alone. Let

$$\theta = \theta(J), \quad \theta'(J) > 0, \quad \theta(1) = 0, \quad \theta'(1) = 1, \quad (3)$$

be a strictly increasing volumetric strain measure, normalized to vanish with unit slope in the undeformed state. The normalization has two consequences. It makes κ in (5) the physical bulk modulus for every admissible θ , so that two instances of the formulation are calibrated to the same material rather than to two different ones. And since $\partial J / \partial \mathbf{F} = J \mathbf{F}^{-\top}$,

$$\frac{\partial \theta}{\partial \mathbf{F}} = \theta'(J) J \mathbf{F}^{-\top}, \quad (4)$$

so (3) also forces $\partial \theta / \partial \mathbf{F} = \mathbf{I}$ at $\mathbf{F} = \mathbf{I}$ — which is what makes the pressure–displacement pairing reduce to $\int \bar{p} \text{Div } \boldsymbol{\psi} \, dV$ independently of the choice of θ (§3.1).

Two choices are used below and are developed in §2.6: the logarithmic measure $\theta = \log J$, and $\theta = J - 1$. Both satisfy (3), and by Remark 2 they nevertheless give different elements.

2.2 Stored energy

The formulation is developed for a stored energy that splits exactly along (2), with a volumetric part quadratic in θ :

$$W(\mathbf{F}) = W^{\text{vol}}(\theta) + W^{\text{iso}}(\bar{\mathbf{F}}), \quad W^{\text{vol}}(\theta) = \frac{\kappa}{2} \theta^2. \quad (5)$$

Both halves of (5) are requirements on the material rather than conventions, they are requirements of different strength, and §2.5 separates them and reports which materials meet them.

Given (5), the volumetric constitutive relation is linear,

$$p = \partial W^{\text{vol}} / \partial \theta = \kappa \theta, \quad (6)$$

and it is this linearity — not the choice of θ , and not the logarithmic measure specifically — that makes the discrete pressure equal $\kappa \bar{\theta}$ pointwise in §3.2. The elimination itself rests on less, as §2.5 shows.

The isochoric response is whatever W^{iso} the material supplies. It is evaluated at the pointwise $\bar{\mathbf{F}}$ and is never projected, which is the property §4 requires. Plasticity acts entirely inside W^{iso} , through an isochoric internal variable evolved by a return map; neither (5) nor (6) changes when it does.

2.3 Three-field functional

Only the volumetric response is treated mixedly. Introduce an independent volumetric strain $\bar{\theta}$ and constrain it to $\theta(\boldsymbol{\varphi})$ through a multiplier $\bar{p}$, which at stationarity is the pressure:

$$\begin{aligned} \Pi[\boldsymbol{\varphi}, \bar{\theta}, \bar{p}] := & \int_{\Omega} W^{\text{vol}}(\bar{\theta}) \, dV + \int_{\Omega} W^{\text{iso}}(\bar{\mathbf{F}}(\boldsymbol{\varphi})) \, dV + \int_{\Omega} \bar{p} (\theta(\boldsymbol{\varphi}) - \bar{\theta}) \, dV \\ & - \int_{\Omega} \rho_0 \mathbf{b} \cdot \boldsymbol{\varphi} \, dV - \int_{\partial_T \Omega} \mathbf{T} \cdot \boldsymbol{\varphi} \, dS, \end{aligned} \quad (7)$$

with ρ_0 the reference density, $\mathbf{b}$ the body force per unit mass, and $\mathbf{T}$ the traction prescribed on the part $\partial_T \Omega$ of the boundary. The isochoric energy is evaluated at the pointwise $\bar{\mathbf{F}}(\boldsymbol{\varphi})$ — for the Hencky instance of §2.6, at $\mathbf{e}(\boldsymbol{\varphi})$ — and carries no auxiliary field; Remark 4 gives the reason.

2.4 What the functional reduces to

With $\bar{\theta}$ and $\bar{p}$ in one discrete space, (7) reduces to the mean-dilatation formulation of Simo et al. [16]; this section gives the reduction.

Put θ and $\bar{p}$ in the same discrete space Q_h , as §3.1 requires. Stationarity in $\bar{\theta}$ then gives $\bar{p} = \kappa \bar{\theta}$ exactly rather than after a projection, and stationarity in $\bar{p}$ gives $\bar{\theta} = \mathcal{P}_h \theta$, the L^2 projection of θ onto Q_h . The constraint term of (7) then vanishes identically:

$$\int_{\Omega} \bar{p} (\theta - \bar{\theta}) \, dV = \kappa \int_{\Omega} \bar{\theta} (\theta - \mathcal{P}_h \theta) \, dV = 0, \quad (8)$$

because $\kappa \bar{\theta} \in Q_h$ while $(\theta - \mathcal{P}_h \theta) \perp Q_h$ by the definition of the projection. What is left is a functional in the motion alone,

$$\Pi[\boldsymbol{\varphi}] = \int_{\Omega} W^{\text{iso}}(\bar{\mathbf{F}}(\boldsymbol{\varphi})) \, dV + \int_{\Omega} \frac{\kappa}{2} (\mathcal{P}_h \theta)^2 \, dV, \quad (9)$$

whose stationarity conditions are exactly the three-field equations of §2.7.

Equation (9) is the mean-dilatation formulation of Simo et al. [16], in which the volumetric strain is replaced by its projection onto an element-wise space, and the mechanism of the

$\bar{B}$ family (projected strain–displacement operator) and the $\bar{F}$ family (projected deformation gradient) of elements; Boffi et al. [3, §8.12] set out the relation between reduced integration, projected pressure, and the mixed method it is equivalent to. The reduction is exact by (8), and no property of θ or of W^{vol} enters it: the exactness is a property of the discretization, that the two auxiliary fields share one space. What the material must supply is stated in §2.5, and the quantity the formulation leaves open is Q_h , which §3.1 selects and §8 measures.

2.5 What the formulation requires of the material

Section 2.4 shows the reduction is exact whenever $\bar{p}$ and $\bar{\theta}$ share a space. That is a statement about the discretization and says nothing about the material. What the material must supply is (5), and the two halves of it are requirements of different strength. They are separated below, together with a third condition that is not a requirement of this section.

An exact volumetric–isochoric split. This condition is structural. The element evaluates W at the pointwise deformation, subtracts the pointwise volumetric energy and adds it back at the projected strain, so the subtraction must leave no volumetric remainder:

$$W(\mathbf{F}) = W^{\text{vol}}(J) + W^{\text{iso}}(\bar{\mathbf{F}}), \quad \bar{\mathbf{F}} := J^{-1/3}\mathbf{F}, \quad \det \bar{\mathbf{F}} = 1. \quad (10)$$

Whether a material satisfies (10) is decidable without knowing W^{vol} : if the split exists then $W(\mathbf{F}) - W(\bar{\mathbf{F}}) = W^{\text{vol}}(J) - W^{\text{vol}}(1)$ depends on J alone, so holding J fixed while varying the isochoric part must leave the difference unmoved. Column 2 of Table 1 is that test, run over every material implemented in Norma and Carina, the two finite-element codes in which the formulation is to be implemented. Five of the seven satisfy it; the element-level implementation in Norma admits only Hencky.

A quadratic W^{vol} . This condition is a convenience and not a necessity. Stationarity in $\bar{p}$ gives $\bar{\theta} = \mathcal{P}_h\theta$ regardless of the material, and stationarity in $\bar{\theta}$ then gives

$$\bar{p} = \mathcal{P}_h\left(\partial W^{\text{vol}}/\partial \bar{\theta}\right) \quad (11)$$

in general — the projection of a known function of the already computed $\bar{\theta}$, one more mass-matrix solve with a nonlinear integrand and nothing implicit. When W^{vol} is quadratic its derivative is linear, $\kappa\bar{\theta}$ already lies in Q_h , the projection in (11) is the identity, and $\bar{p} = \kappa\bar{\theta}$ holds pointwise; otherwise $\bar{p} = \mathcal{P}_h W^{\text{vol}'}(\mathcal{P}_h\theta)$ is a doubly projected pressure. The reduction (9) is exact either way, with $W^{\text{vol}}(\mathcal{P}_h\theta)$ in place of the quadratic, and the elimination of §3.2 is linear either way; what the quadratic case gives is the pointwise relation and one projection fewer.

The logarithmic measure itself. This condition is required by nothing in the present section. It is required by §4, for an unrelated reason: only in logarithmic strain is the constitutive law algebraically linear, which is what allows a small-strain return map to be reused verbatim.

Table 1: What the materials implemented in Norma satisfy. Column 2 is the spread of $W(\mathbf{F}) - W(\bar{\mathbf{F}})$ across twenty random isochoric parts at each of four fixed $J \in [0.85, 1.2]$, relative to its own magnitude; a spread below 10^{-12} means the split (10) holds to rounding error. Columns 3 and 4 are the relative residual of a least-squares fit of the extracted $W^{\text{vol}}(J) = W(J^{1/3}\mathbf{I}) - W(\mathbf{I})$ over $J \in [0.8, 1.25]$ to a quadratic in each candidate volumetric variable; in both exact cases the fitted coefficient is $\kappa/2$ to five digits. Two materials admit a quadratic W^{vol} , and they disagree on the variable. Script: `prototype/materials.jl`, which needs Norma.

material	split (10)	W^{vol} quadratic in	
	spread at fixed J	$J - 1$	$\log J$
Linear elastic	2.0×10^0	—	—
Saint-Venant Kirchhoff	3.0×10^1	—	—
Neo-Hookean	4.2×10^{-13}	5.9×10^{-2}	1.3×10^{-1}
Reciprocal neo-Hookean	4.8×10^{-13}	1.8×10^{-1}	6.5×10^{-3}
Seth–Hill ($m = n = 1$)	4.3×10^{-13}	1.8×10^{-1}	6.5×10^{-3}
Hencky	1.4×10^{-13}	1.8×10^{-1}	1.7×10^{-15}
J_2 , Simo–Hughes	5.9×10^{-13}	5.3×10^{-15}	1.8×10^{-1}

The case developed here. The formulation is developed in detail for materials satisfying (10) with a quadratic W^{vol} , because that is where the pointwise relation holds and what both implemented materials satisfy. Table 1 admits exactly two, and they do not agree on the variable: Hencky is quadratic in $\log J$ to 1.7×10^{-15} , and the Simo–Hughes J_2 model that Norma and Carina both implement is quadratic in $J - 1$ to 5.3×10^{-15} . Each is exact in its own variable and wrong by 18% in the other, so material and volumetric variable are not independently selectable, and neither choice generalizes to the other material. §2.6 develops both.

The three remaining split materials — neo-Hookean, its reciprocal, and Seth–Hill — are admissible with one projection more (Remark 1) and are not developed here. What the formulation requires of a material is the split, and five of the seven have it.

Remark 1 (Extending to a non-quadratic W^{vol}). Neither §2.3 nor §2.4 is affected, and no local Newton iteration is introduced. Two things change.

First, (19) acquires a second projection: $\bar{\theta} = \mathcal{P}_h \theta$ from one mass-matrix solve as before, then $\bar{p} = \mathcal{P}_h W^{\text{vol}'}(\bar{\theta})$ from a second solve with the same factorization and a nonlinear integrand. The tangent contribution is $\int W^{\text{vol}''}(\bar{\theta}) \mathcal{P}_h(D\theta[\delta\varphi]) \mathcal{P}_h(D\theta[\psi]) \, dV$, a weighted integral against the same constant mass matrix, and the factorization is reused across Newton iterations exactly as in the quadratic case.

Second, the choice of volumetric variable is not determined by the material. For a quadratic W^{vol} the variable is whichever one makes it quadratic, and Table 1 settles it. For a non-quadratic W^{vol} any monotone $\theta(J)$ satisfying (3) is admissible, and by Remark 2 the choice changes the element: the choice of θ changes the discrete element and not only the computation.

Remark 2 (The volumetric variable is a modeling choice). Projecting $\log J$ and projecting $J - 1$ do not give the same element. Measured on one tetrahedron under a displacement quadratic in $\mathbf{X}$, with the element mean as the projection and a Hencky material so that both variants use the same exact W^{vol} , the element energies differ by 9.4×10^{-4} at 0.5% nominal strain,

Table 2: The two instances of (5). Everything above the rule is fixed by the generic development; everything below it is where the instances differ. $\mathbf{b}^e := \mathbf{F}^e(\mathbf{F}^e)^\top$ is the elastic left Cauchy–Green tensor, with $\mathbf{F}^e$ the elastic part of the multiplicative decomposition $\mathbf{F} = \mathbf{F}^e \mathbf{F}^p$, and $\bar{\mathbf{b}}^e := J^{-2/3} \mathbf{b}^e$ its isochoric part.

	Hencky	Simo–Hughes
volumetric variable θ	$\log J$	$J - 1$
$\theta'(J)$	$1/J$	1
$\partial\theta/\partial\mathbf{F}$	$\mathbf{F}^{-\top}$	$J \mathbf{F}^{-\top}$
$W^{\text{vol}}(\theta)$	$\frac{\kappa}{2}(\log J)^2$	$\frac{\kappa}{2}(J - 1)^2$
isochoric argument	$\mathbf{e} = \text{dev } \boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} = \frac{1}{2} \log \mathbf{C}$	$\bar{\mathbf{b}}^e$
W^{iso}	$\mu \ \mathbf{e}\ ^2$	$\frac{\mu}{2}(\text{tr } \bar{\mathbf{b}}^e - 3)$
return map	radial return in $\boldsymbol{\epsilon}$	radial return on $\bar{\mathbf{b}}^e$
small-strain models usable unmodified	yes	no
implemented in Norma and Carina	elasticity only	yes, with consistent tangent

1.0×10^{-2} at 5%, 5.6×10^{-2} at 20% and 2.7×10^{-1} at 50% (`prototype/materials.jl`). The difference is $O(\text{strain})$ and vanishes with it, so both are consistent discretizations of the same continuum problem; they are different elements, and they differ by tens of percent in exactly the finite-deformation regime this formulation exists to address. Which variable is projected is therefore a modeling decision, not a change of notation.

2.6 The two admissible instances

Table 1 leaves two materials. They agree on everything §2.1–§2.4 requires and differ in every quantity of the formulation that depends on the instance; Table 2 collects both columns.

The Hencky instance is the one (12) defines: with

$$\boldsymbol{\epsilon} := \frac{1}{2} \log \mathbf{C} \in \mathbb{S}, \quad \mathbf{C} := \mathbf{F}^\top \mathbf{F}, \quad (12)$$

evaluated by inverse scaling and squaring with Padé approximants — repeated square roots of $\mathbf{C}$ followed by a rational approximant of the logarithm [2] — the volumetric–deviatoric decomposition

$$\boldsymbol{\epsilon} = \frac{1}{3} \theta \mathbf{I} + \mathbf{e}, \quad \theta = \text{tr } \boldsymbol{\epsilon} = \log J, \quad \mathbf{e} = \text{dev } \boldsymbol{\epsilon}, \quad (13)$$

is orthogonal in the Frobenius inner product, and the split of (5) is inherited from the strain measure rather than imposed on it. The Simo–Hughes instance imposes it instead, through the $J^{-2/3}$ factor in $\bar{\mathbf{b}}^e$; $\det \bar{\mathbf{b}}^e = 1$ then makes W^{iso} independent of J by construction. Both are exact, and Table 1 confirms both to machine precision. Three consequences follow.

The generic development covers both. Nothing in §2.1–§2.4, and nothing in the discretization of §3.1–§3.2, refers to either column of Table 2. The reduction (9), the pointwise relation $\bar{p} = \kappa \theta$, and the element-local elimination hold for any θ satisfying (3) and any W^{iso} satisfying (5).

The pairing at $\mathbf{F} = \mathbf{I}$ is instance-independent, and §8 relies on this. By (4) and the normalization (3), $\partial\theta/\partial\mathbf{F} = \mathbf{I}$ at $\mathbf{F} = \mathbf{I}$ in both columns, so the pairing that §3.1 reduces to $\int \bar{p} \operatorname{Div} \boldsymbol{\psi} \, dV$ is the same for both. The inf-sup analysis, and with it the negative stability results of §8.5, therefore transfer between instances unchanged: those results are a property of the pair (U_h, Q_h) and do not depend on the instance on which the measurement is performed.

The instances are not interchangeable in plasticity, and this is the only place they are not. The last two rows of Table 2 are the entire difference, and they follow from (13): only in the logarithmic instance is the constitutive law algebraically linear in the strain that enters the return map, which is what lets a small-strain algorithm be reused without reformulation. §4 develops that instance. Remark 6 records that it is the other one that both codes implement.

2.7 Variations

With $\boldsymbol{\varphi} \in \mathcal{U}$, $\bar{\theta}, \bar{p} \in \mathcal{Q}$, and test functions $\boldsymbol{\psi} \in \mathcal{U}$ vanishing on $\partial\boldsymbol{\varphi}\Omega$, the part of the boundary on which the motion is prescribed, and $\vartheta, \pi \in \mathcal{Q}$:

$$\begin{aligned} D_{\boldsymbol{\varphi}}\Pi(\boldsymbol{\psi}) &= \int_{\Omega} \bar{p} \frac{\partial\theta}{\partial\boldsymbol{\varphi}}[\boldsymbol{\psi}] \, dV + \int_{\Omega} \mathbf{s}(\mathbf{e}(\boldsymbol{\varphi})) : \frac{\partial\mathbf{e}}{\partial\boldsymbol{\varphi}}[\boldsymbol{\psi}] \, dV - \int_{\Omega} \rho_0 \mathbf{b} \cdot \boldsymbol{\psi} \, dV - \int_{\partial_T\Omega} \mathbf{T} \cdot \boldsymbol{\psi} \, dS = 0, \\ D_{\bar{\theta}}\Pi(\vartheta) &= \int_{\Omega} (\kappa\bar{\theta} - \bar{p}) \vartheta \, dV = 0, \\ D_{\bar{p}}\Pi(\pi) &= \int_{\Omega} (\theta(\boldsymbol{\varphi}) - \bar{\theta}) \pi \, dV = 0. \end{aligned} \tag{14}$$

The kinematic derivatives are

$$\frac{\partial\theta}{\partial\boldsymbol{\varphi}}[\boldsymbol{\psi}] = \theta'(J) J \mathbf{F}^{-\top} : \operatorname{Grad} \boldsymbol{\psi}, \quad \frac{\partial\mathbf{e}}{\partial\boldsymbol{\varphi}}[\boldsymbol{\psi}] = \mathbb{L}^{\operatorname{dev}}(\mathbf{C}) : \operatorname{sym}(\mathbf{F}^{\top} \operatorname{Grad} \boldsymbol{\psi}), \tag{15}$$

with $\mathbb{L}(\mathbf{C}) := \partial \log \mathbf{C} / \partial \mathbf{C}$ the Fréchet derivative of the matrix logarithm [2] and $\mathbb{L}^{\operatorname{dev}} := \mathbb{L} - \frac{1}{3} \mathbf{I} \otimes \mathbf{C}^{-1}$. The volumetric derivative follows from Jacobi's formula and (4). The deviatoric expression is written for the Hencky instance of Table 2; the Simo–Hughes instance replaces it with the variation of $\bar{\mathbf{b}}^e$, and nothing outside this equation changes.

2.8 Stress reconstruction

Collecting (14)₁ into the form $\int \mathbf{P} : \operatorname{Grad} \boldsymbol{\psi}$ gives, in general,

$$\mathbf{P} = \theta'(J) J \bar{p} \mathbf{F}^{-\top} + \mathbf{F} (\mathbb{L}(\mathbf{C}) : \mathbf{s}). \tag{16}$$

The deviatoric term is $\mathbb{L}$ and not $\mathbb{L}^{\operatorname{dev}}$ applied to $\mathbf{s}$. Moving $\mathbb{L}^{\operatorname{dev}}$ off the strain variation in (15) and onto $\mathbf{s}$ takes its transpose, which carries the $\frac{1}{3}$ term onto $\operatorname{tr} \mathbf{s} = 0$, and $\mathbb{L}$ is self-adjoint because $\log \mathbf{C}$ is the gradient of $\operatorname{tr}(\mathbf{C} \log \mathbf{C} - \mathbf{C})$. Writing $\mathbb{L}^{\operatorname{dev}} : \mathbf{s}$ in its place adds $-\frac{1}{3}(\mathbf{C}^{-1} : \mathbf{s})\mathbf{F}$, which is not zero, since $\operatorname{tr} \mathbf{s} = 0$ does not make $\mathbf{C}^{-1} : \mathbf{s}$ vanish; the error is invisible in elasticity, where (16) collapses to (17).

When the deviatoric response is elastic, $\mathbf{s} = 2\mu \mathbf{e}$ is a function of the current $\mathbf{C}$ alone through an isotropic law, so $\mathbf{s}$ is coaxial with $\mathbf{C}$ at every quadrature point; the off-diagonal blocks of $\mathbb{L}$ in the principal basis then act on nothing and (16) collapses to

$$\mathbf{P} = (\varpi \mathbf{I} + \mathbf{s}_x) \mathbf{F}^{-\top} = \boldsymbol{\tau} \mathbf{F}^{-\top}, \quad \boldsymbol{\tau} := \varpi \mathbf{I} + \mathbf{s}_x = J \boldsymbol{\sigma}, \quad \varpi := \theta'(J) J \bar{p}, \tag{17}$$

in which $\boldsymbol{\tau} = J \boldsymbol{\sigma}$ is the Kirchhoff stress, $\boldsymbol{\sigma}$ the Cauchy stress, and $\boldsymbol{s}_x := \boldsymbol{R} \boldsymbol{s} \boldsymbol{R}^\top$ the spatial deviator, with $\boldsymbol{R}$ the rotation of the polar decomposition $\boldsymbol{F} = \boldsymbol{R} \boldsymbol{U}$. The material $\boldsymbol{s}$ of (13) and the spatial $\boldsymbol{s}_x$ are different tensors; in material form the same statement reads $\boldsymbol{P} = \boldsymbol{F}^{-\top} \boldsymbol{s}$, which agrees with (17) because $\boldsymbol{s}$ and $\boldsymbol{U}$ commute when coaxial.

The Kirchhoff pressure ϖ carries the whole of the instance dependence. For the Hencky instance $\theta'(J)J = 1$ and $\varpi = \bar{p}$; for the Simo–Hughes instance $\theta'(J)J = J$ and $\varpi = J\bar{p}$, giving $\boldsymbol{\tau} = J\bar{p} \boldsymbol{I} + \boldsymbol{s}$ with $\bar{p} = \kappa(J - 1)$, which is the Kirchhoff stress of Simo and Hughes [15, BOX 9.1] and is what both codes compute. Agreement of (16) with both implementations confirms the normalization (3).

Remark 3 (Coaxiality is an elastic property). Equation (17) rests on $\boldsymbol{s}$ being coaxial with $\boldsymbol{C}$, which holds only when $\boldsymbol{s}$ is a function of the current $\boldsymbol{C}$ alone. It fails in two settings, and the second is the one this formulation exists for. With an independent discrete stress field, $\boldsymbol{s}$ is tied to $\boldsymbol{C}$ only weakly and is not coaxial with it at an arbitrary iterate. With plasticity, the return map (20) evaluates $\boldsymbol{s} = 2\mu(\boldsymbol{e} - \boldsymbol{e}_{n+1}^p)$ against a stored plastic strain $\boldsymbol{e}_n^p$, which is coaxial with the current $\boldsymbol{C}$ only under proportional loading; whenever the principal axes rotate — the large-rotation regime of §1 — $\boldsymbol{s}$ is not coaxial with $\boldsymbol{C}$, the off-diagonal blocks of $\mathbb{L}$ do not vanish, and (17) is wrong. In both settings (17) holds at best at a converged proportional state, and assembling the residual from it gives a residual that is not the derivative of the energy, with an error that is undetected under proportional loading. Equation (16) is the form to implement, and Miehe et al. [9] carry the full projection tensor for exactly this reason.

Remark 4. No deviatoric mixed pair appears because none is needed. Volumetric locking is a constraint phenomenon: at $\kappa/\mu \gg 1$ the near-isochoric constraint over-determines the displacement space, and relieving it requires a multiplier. The deviatoric response is not constrained in the same sense and admits no analogous locking mode. Adding a deviatoric pair adds degrees of freedom, an inf-sup condition that has to be established separately, and — if strain and stress are placed in different spaces — a constitutive law evaluated across two discretizations, which is the error that corrupts plastic strain fields when the strain is interpolated in one space and the stress evaluated in another. §7 gives the construction that adds strain as an independent field should a later requirement demand it.

3 Discretization

This section selects the pair of discrete spaces and gives the element-local elimination of the auxiliary fields. P_k denotes continuous piecewise polynomials of degree k on the mesh, P_k^{disc} the discontinuous ones, and a bubble a polynomial on one element that vanishes on a prescribed part of its boundary; U_h/Q_h denotes the pair with displacement space U_h and pressure space Q_h .

3.1 The pair

Let $U_h \subset \mathcal{U}$ carry the motion and $Q_h \subset \mathcal{Q}$ carry both $\bar{\theta}$ and $\bar{p}$. Two requirements fix the choice.

Same space for both scalars. $\bar{\theta}$ and $\bar{p}$ are placed in one space Q_h . This is what makes the constraint term vanish identically and the functional reduce to (9); it is the only thing that reduction needs. Given it, and a quadratic W^{vol} (§2.5), (14)₂ reads $\bar{p} = \kappa \bar{\theta}$ pointwise

rather than in a projected sense, and the composition $\bar{p} = \kappa \mathcal{P}_h(\theta)$ is well defined. Splitting the two scalars across different spaces makes the constitutive relation hold only after a projection, determines one field only through its projection onto the other's space, and can leave the discrete system singular.

Inf-sup stability, in three dimensions. By (15) the pressure–displacement pairing is $\int \varpi \mathbf{F}^{-\top} : \text{Grad } \boldsymbol{\psi}$ with $\varpi = \theta'(J) J \bar{p}$, and not $\int \bar{p} \mathbf{F}^{-\top} : \text{Grad } \boldsymbol{\psi}$; the two coincide at $\mathbf{F} = \mathbf{I}$ and nowhere else. The normalization (3) sets $\theta'(1) = 1$, so $\varpi = \bar{p}$ there and the pairing reduces to the divergence pairing $\int \bar{p} \text{Div } \boldsymbol{\psi} \, dV$ — in both instances of Table 2, which is what makes the stability results of §8.5 a property of the pair rather than of the material. That pairing is the one of incompressible linear elasticity, in which the pressure is the multiplier of the constraint $\text{Div } \boldsymbol{\varphi} = 0$, and its discrete inf-sup condition,

$$\inf_{q \in Q_h} \sup_{\mathbf{v} \in U_h} \frac{\int_{\Omega} q \, \text{Div } \mathbf{v} \, dV}{\|\mathbf{v}\|_{H^1} \|q\|_{L^2}} \geq \beta_h, \quad (18)$$

is the condition on the pair. The admissible pairs are therefore those for which β_h in (18) is bounded below uniformly in h , and stability must also hold on the target range of κ/μ [3].

Stability of a pair depends on the spatial dimension. The conforming Crouzeix–Raviart pair [4] is, in two dimensions, P_2 enriched with a cubic interior bubble over P_1^{disc} ; its three-dimensional counterpart below carries a quartic interior bubble and cubic face bubbles, and the quadrature degree the element needs follows from the quartic. Unenriched P_2/P_1^{disc} is stable in two dimensions only on special meshes. In three dimensions Boffi et al. [3, §8.7] state that even a

constant discontinuous pressure is not controlled by P_2 : “to control piecewise constant pressure, we need some degrees of freedom on the faces,” and the quadratic space has none. The stable three-dimensional choices with a discontinuous pressure therefore require bubble enrichment of the displacement space, and the enrichment has two parts with two functions: face bubbles control the constant pressure, and an interior bubble lifts it to a linear one [3, Example 8.7.2]. Table 3 lists the candidates, and §8.5 measures every one of them.

Displacement	Pressure	3D stable	Pressure elimination
P_1	P_0	no	local
P_2	P_0	no	local
$P_2 \oplus b_{\text{face}}$	P_0	yes	local
P_2	P_1^{disc}	no	local
$P_2 \oplus b_{\text{int}}$	P_1^{disc}	no	local
$P_2 \oplus b_{\text{int}} \oplus b_{\text{face}}$	P_1^{disc}	yes	local
P_2	P_1 continuous	yes	global

Table 3: Displacement–pressure pairs on tetrahedra, with three-dimensional stability as Boffi et al. [3, §8.7, Example 8.7.2] give it. Local elimination requires a discontinuous pressure; three-dimensional stability with a discontinuous pressure requires face bubbles, and a linear pressure requires an interior bubble as well.

The pair this formulation requires is P_2 enriched with face and interior bubbles over P_1^{disc} . The two enrichments are different kinds of object. The interior bubble vanishes on the whole element boundary: it is never shared, never reached by a boundary condition, and condenses away, adding no global unknown. A face bubble vanishes on three faces and not on its own, so the two elements meeting there share it; it is a global unknown and does not condense. The enrichment therefore adds global degrees of freedom, which §8.3 counts. The pressure elimination of §3.2 nevertheless stays element by element, because it rests on Q_h being discontinuous and not on the displacement space.

The linear pressure is not forced by the stability results. $P_2 \oplus b_{\text{face}}/P_0$ is also stable (§8.5), at a higher constant and with the same number of global unknowns, since the interior bubble condenses. The count of §8.2 applies to the unenriched displacement space, and §8.4 shows that its modes do not persist after assembly. What separates the two pressure spaces is the claim of §5: a constant pressure leaves in $\ker \mathbf{G}_h$ the element-scale dilatational modes that a plastic tangent can free, and a linear one does not. §8.7 measures both pairs, and that measurement is what the linear pressure is for.

Both that pair and the Taylor–Hood pair — continuous P_2 over continuous P_1 , whose stability in three dimensions is established [3] and which serves as the positive control of the measurements — are measured in §8.5, and every other element-local pair decays. The difference between the two stable pairs is not stability but where the coupling is — a global pressure that cannot be eliminated element by element against global face bubbles that cannot be condensed — and §10 quantifies it.

3.2 Local elimination

Because Q_h is discontinuous, the mass matrix $[M]_{mn} = \int_{\Omega} \chi_m \chi_n \, dV$, with $\{\chi_m\}$ a basis of Q_h , is block diagonal by element and constant in the reference configuration. One factorization at setup serves every Newton iteration of every step, and

$$M\{\bar{\theta}\} = \int_{\Omega} \chi_m \theta(\varphi) \, dV, \quad \bar{p} = \kappa \bar{\theta}, \quad (19)$$

is an element-local solve, with $\theta = \log J$ for a Hencky material and $\theta = J - 1$ for the Simo–Hughes one (§2.5). The second relation in (19) holds pointwise, rather than only after projection, precisely because W^{vol} is quadratic; for a material where it is not, it is replaced by a second projection with the same factorization (Remark 1). Substituting $\bar{p}(\varphi)$ into (14)₁ leaves a displacement-sized nonlinear problem whose tangent retains the sparsity of the displacement mesh — the auxiliary fields never couple two elements. This is the property that keeps the composite tetrahedron’s tangent as sparse as a displacement element’s, and it is preserved here.

Remark 5. The locality is a consequence of the discontinuous pressure, not of the Hu–Washizu structure. A conforming auxiliary field — one requiring inter-element continuity — would make $\bar{p}(\varphi)$ a global function of φ , and every tangent–vector product would then require a global solve with its mass matrix, which the composite tetrahedron does not require. That is the main reason the pressure is kept discontinuous.

4 Plasticity

This section develops the logarithmic instance of Table 2. It is the only part of the formulation where the choice of instance changes anything, and the reason to prefer this one; what follows does not hold verbatim for the Simo–Hughes column, and Remark 6 records where that leaves the verification program.

The deviatoric response is evaluated at quadrature points of the displacement element, on $e(\varphi)$ computed there. The plastic strain is carried as a logarithmic strain e^p and subtracted additively — the construction of Miehe et al. [9], in which a material geometric pre-processing $\mathbf{C} \mapsto \epsilon$ and a post-processing through $\mathbb{L}(\mathbf{C})$ precede and follow an unmodified small-strain algorithm. This is a different constitutive model from the multiplicative $\mathbf{F} = \mathbf{F}^e \mathbf{F}^p$ plasticity of Simo [14], which also preserves the small-strain return map but applies it to the elastic logarithmic strain of a trial elastic state with an exponential update; the two coincide under proportional loading with fixed principal axes and differ when the axes rotate. Because the constitutive law is linear in logarithmic strain, the J_2 return map is the small-strain radial return verbatim:

$$\begin{aligned} e^{\text{tr}} &:= e(\varphi) - e_n^p, & s^{\text{tr}} &:= 2\mu e^{\text{tr}}, \\ \Delta\gamma : \quad & \|s^{\text{tr}}\| - 2\mu \Delta\gamma - \sqrt{2/3} \left[\sigma_y + \mathcal{H}'(\bar{\epsilon}_n^p + \sqrt{2/3} \Delta\gamma) \right] = 0 & \text{if } f^{\text{tr}} > 0, \quad \text{else } \Delta\gamma := 0, \\ e_{n+1}^p &:= e_n^p + \Delta\gamma n, & n &:= s^{\text{tr}} / \|s^{\text{tr}}\|, & \bar{\epsilon}_{n+1}^p &:= \bar{\epsilon}_n^p + \sqrt{2/3} \Delta\gamma, \\ s &:= 2\mu (e(\varphi) - e_{n+1}^p). \end{aligned} \tag{20}$$

Here σ_y is the initial yield stress, $\bar{\epsilon}^p$ the equivalent plastic strain, $\mathcal{H}'$ the hardening function (the increase of the yield stress with $\bar{\epsilon}^p$) and $\mathcal{H}''$ its derivative, the hardening modulus, and $f^{\text{tr}} := \|s^{\text{tr}}\| - \sqrt{2/3} [\sigma_y + \mathcal{H}'(\bar{\epsilon}_n^p)]$ the trial yield function. The consistency condition is a scalar equation in $\Delta\gamma$. For linear hardening it closes in one step, $\Delta\gamma = f^{\text{tr}} / (2\mu + \frac{2}{3}\mathcal{H}'')$, which is the form both codes implement; for nonlinear hardening it is a scalar Newton iteration [15, BOX 3.2]. Internal variables are stored at quadrature points and are never projected, so no interpolation error enters the history. Frame-indifferent interpolation, when tensorial internal variables must be transferred, follows Mota et al. [10].

Three properties follow. They are the motivation for the formulation rather than incidental consequences of it.

The constitutive law is untouched. All finite-deformation content is absorbed into two material-independent geometric transforms: the forward map $\mathbf{C} \mapsto \epsilon$ and the tangent map $\mathbb{L}(\mathbf{C})$ [9]. Between them sits an unmodified small-strain algorithm. Any model already written for small strain — J_2 , kinematic hardening, viscoplasticity, damage — is usable without reformulation. This is the practical content of preferring a Hu–Washizu structure over one built around a particular stress–strain pairing. Note that this property belongs to the logarithmic measure and not to the Hu–Washizu structure, so it is the one place where §2.5 does not apply: an admissible material in the sense of (10) need not have it, and the Simo–Hughes model both codes implement does not (Remark 6).

Strain and stress never cross discretizations. $\mathbf{e}$ is evaluated and $\mathbf{s}$ returned at the same quadrature point of the same element. There is no projection between the two, so the aliasing that corrupts equivalent plastic strain in stabilized formulations has no mechanism here.

No stabilization enters the deviator. There is no subgrid term, no penalty and no tuned coefficient anywhere in the formulation, so nothing of the form (1) can reach the return map.

Remark 6 (The model implemented in both codes, and its bearing on claim 4). Equation (20) is the return map this formulation specifies. Neither Norma nor Carina implements it. Both implement the Simo–Hughes J_2 model [15]: a neo-Hookean volumetric–isochoric split, $W = \frac{\kappa}{2}(J-1)^2 + \frac{\mu}{2}(\text{tr } \bar{\mathbf{b}}^e - 3)$, with the radial return taken on the isochoric elastic left Cauchy–Green tensor rather than on logarithmic strain. It satisfies (10) exactly — $\det \bar{\mathbf{b}}^e = 1$ makes the isochoric term independent of J by construction rather than by an identity — and its W^{vol} is quadratic in $J-1$, so it is admissible here in the variable $J-1$.

Two consequences follow. The first is that claim 4 of §9 will be measured on Simo–Hughes unless a logarithmic J_2 is written for the purpose, and the substitution is larger than an elastic split: it exchanges the additive logarithmic model of (20) for multiplicative plasticity with the return map on $\bar{\mathbf{b}}^e$. The smoothness of equivalent plastic strain is a question about whether strain and stress cross discretizations, which is independent of the plasticity model carrying them; the substitution changes the plasticity model and not only the volumetric split.

The second is that the argument above that small-strain models are usable unmodified is weakened by it. That property belongs to the logarithmic measure, and Simo–Hughes does not have it: its return map is specific to its own structure. The argument stands for the formulation as specified and does not describe what is implemented.

5 The claim

In elasticity the second hypothesis holds for every conforming pair: by Korn’s inequality, which bounds $\int |\text{Grad } \mathbf{v}|^2 dV$ by a multiple of $\int |\text{dev } \boldsymbol{\epsilon}(\mathbf{v})|^2 dV$ for $\mathbf{v} \in H_0^1$, the deviatoric form is coercive on all of U_h , so it is coercive on $\ker \mathbf{G}_h$, the set of discrete displacements whose divergence has zero projection onto Q_h (§8.1), and no elastic soft mode is a failure of the theorem. A count of exact zero-energy modes on one element is therefore not an argument against a constant pressure in elasticity; §8.4 shows that those modes do not persist after assembly.

Under plastic loading the consistent J_2 tangent, the derivative of the stress returned by (20) with respect to the strain, is

$$\mathbb{C}^{\text{ep}} = 2\mu \left(\mathbb{I}^{\text{dev}} - \eta \mathbf{n} \otimes \mathbf{n} \right), \quad \mathbf{n} := \mathbf{s}^{\text{tr}} / \|\mathbf{s}^{\text{tr}}\|, \quad \eta = \frac{2\mu}{2\mu + \frac{2}{3}\mathcal{H}''}, \quad (21)$$

with $\eta = 1$ for perfect plasticity: the shear stiffness along the flow direction is removed.

Read as a constant, the second hypothesis cannot separate pairs. For every $\mathbf{v} \in H_0^1(\Omega)$, integration by parts gives $\int |\text{dev } \boldsymbol{\epsilon}|^2 dV = \frac{1}{2} \int |\text{Grad } \mathbf{v}|^2 dV + \frac{1}{6} \int (\text{Div } \mathbf{v})^2 dV$, and $(\mathbf{n} : \boldsymbol{\epsilon})^2 \leq |\text{dev } \boldsymbol{\epsilon}|^2$ since $\mathbf{n}$ is a unit deviator, so

$$\frac{1}{2\mu} \int \boldsymbol{\epsilon} : \mathbb{C}^{\text{ep}} : \boldsymbol{\epsilon} dV \geq (1 - \eta) \int |\text{dev } \boldsymbol{\epsilon}|^2 dV \geq \frac{1 - \eta}{2} \int |\text{Grad } \mathbf{v}|^2 dV \quad (22)$$

for every conforming pair and every member of its kernel, isochoric or not. For $\eta < 1$ the coercivity constant on $\ker \mathbf{G}_h$ is therefore bounded below uniformly in h and in the pair. For $\eta = 1$ the continuum operator loses strong ellipticity, positivity of the energy of every displacement jump across a plane: a shear band, a displacement jump $\mathbf{e}_1$ across a plane with normal $\mathbf{e}_2$, has $\epsilon \propto \text{sym}(\mathbf{e}_1 \otimes \mathbf{e}_2)$, to which (21) with that $\mathbf{n}$ assigns zero energy, so the constant tends to zero under refinement for every pair as the discrete space resolves narrower bands (§8.7). Neither limit says anything about the pair.

What distinguishes the pairs is what the soft directions are. A member of $\ker \mathbf{G}_h$ that is nearly isochoric is a shear band the continuum problem also has, and its low energy is physical. A member with pointwise nonzero dilatation of zero element mean has the volumetric energy $\kappa \int (\text{Div } \mathbf{v})^2$ in the continuum and none in the element; once the tangent removes its shear energy it has zero energy in the element and the energy $\kappa \int (\text{Div } \mathbf{v})^2$ in the continuum. That is the soft mode reported for constant-pressure elements, and whether $\ker \mathbf{G}_h$ contains such members is a property of the pair. For a pair satisfying the inf-sup condition, $\ker \mathbf{G}_h$ approximates the space of isochoric displacements at the rate of U_h [3, Prop. 5.1.3], so the dilatation carried by any of its members vanishes with h . For an element-constant pressure on a quadratic displacement, $\ker \mathbf{G}_h$ contains displacements whose dilatation is nonzero pointwise and averages to zero over each element, and their dilatation does not vanish with h .

The claim is therefore the following. *Under a plastic tangent, every soft mode of the stable pair with a linear discontinuous pressure carries a dilatation $\int (\text{Div } \mathbf{v})^2 dV / \int |\text{Grad } \mathbf{v}|^2 dV$ that vanishes under refinement, and a constant pressure has soft modes whose dilatation does not; the soft modes reported for constant-pressure elements are the latter, and no stabilization is needed where they are absent.* Both operators are linear at a fixed plastic state, so the claim is tested without a nonlinear solve, and §8.7 tests it. If the dilatation of the soft modes does not separate the pairs, the formulation has no advantage over the constant-pressure elements it is meant to replace.

6 Dynamics

The spatial formulation is unchanged; the mass matrix enters only the displacement block. Two implementation choices are specified.

Lumping. Row-sum lumping, which assigns to each degree of freedom the row sum of the consistent mass matrix, produces non-positive vertex masses on the quadratic tetrahedron and cannot be used. Diagonal-scaling (HRZ) lumping [6], which scales the diagonal of the consistent mass to preserve the total mass, is positive for any basis, so positivity is not the limitation; on quadratic simplices the known disadvantage of HRZ is accuracy, with the vertex masses small. The limitation the enrichment introduces depends on the basis that carries it, and §8.8 measures both choices. In the hierarchical basis — the ten Lagrange functions of the quadratic tetrahedron together with the five bubbles — a rigid translation has zero coefficient on every bubble degree of freedom, so their diagonal mass is inert for translations, yet HRZ’s normalization counts it: the Lagrange masses sum to less than the total mass by the fraction assigned to the bubbles, 51% on the reference tetrahedron, and a rigid translation carries half its momentum. The normalization must then run over the Lagrange degrees of freedom alone, with each bubble keeping its own consistent diagonal, which restores the momentum exactly. With a nodal basis on the fifteen nodes of the TETRA15 arrangement — vertices, edge midpoints,

face centroids, centroid — the functions form a partition of unity, HRZ conserves momentum without qualification, and the lumped masses are positive on every element measured. Row-sum lumping was also positive in that basis on the three elements measured, unlike for TETRA10, but its positivity is not guaranteed, and HRZ lumping is used.

Critical time step. From the element eigenvalue bound at $\kappa/\mu = 2$ (§8.8), the Courant number $c_p \Delta t/h$, with $c_p = \sqrt{(\kappa + 4\mu/3)/\rho}$ the dilatational wave speed, ρ the density, Δt the stable step and h the element edge length, is 0.48 for TETRA4, 0.20 for TETRA10 under HRZ, and for the enriched element 0.15 with the nodal basis under HRZ and 0.11 with the hierarchical basis under Lagrange-only HRZ; on a Kuhn tetrahedron of the Freudenthal mesh (§8.1), 0.28, 0.13, 0.095 and 0.078. The quadratic element therefore reduces the stable step by a factor of 2.4 against the linear one on the same mesh, and the enrichment a further 1.4 with the nodal basis. A P_2 mesh resolves a given feature with fewer elements; whether the smaller element count compensates for the smaller step depends on the problem and is to be measured.

7 Optional five-field variant

This section gives the extension to an independent deviatoric strain, should one be needed. Should an independent deviatoric strain field become necessary — for non-local regularization, or for internal variables requiring their own discrete representation — the systematic extension introduces $\bar{\mathbf{e}}$ and $\bar{\mathbf{s}}$ in the same discontinuous space, of the same degree as Q_h . Sharing the space is not merely convenient: it keeps the return map at shared quadrature points, keeps every elimination element-local, leaves the discrete system non-singular, and — for a linear deviatoric law, and only then — makes the constitutive relation hold pointwise rather than after a projection; with plasticity $\bar{\mathbf{s}} = \mathcal{P}_h \mathbf{s}(\bar{\mathbf{e}})$ is projected. Placing $\bar{\mathbf{e}}$ and $\bar{\mathbf{s}}$ in different spaces loses each of these properties. Remark 3 also applies: with an independent $\bar{\mathbf{s}}$, the residual must be assembled from (16) and not from (17).

8 Numerical results

Every measurement below is on a linear operator, and all but those of §8.6 are at $\mathbf{F} = \mathbf{I}$: the condensed operator (25) for the count and the locking sweep, the H^1 seminorm for the stability sweep, and the deviatoric form under a plastic tangent for the soft-mode measurements. None needs a solver, a material model, or a mesh reader. The scripts are in `prototype/`, each reports its own result, and `checks.jl` verifies the quadrature, the bubbles, and the conformity of the enriched spaces against values known in advance.

8.1 Method

Meshes are generated rather than read, so the refinement sequence is exact. The unit cube is divided into N^3 cubes and each cube into six tetrahedra by the Freudenthal (Kuhn) subdivision — the six permutations (a, b, c) giving the Kuhn tetrahedra $\{\mathbf{0}, \mathbf{e}_a, \mathbf{e}_a + \mathbf{e}_b, (1, 1, 1)\}$. That decomposition is conforming across shared cube faces, which the five-tetrahedron split is not. Quadratic meshes add edge midpoints in the node order the reference element expects.

For the stability test the displacement is fixed on the entire boundary and three operators are assembled,

$$K_{ij} = \int \text{Grad } N_i : \text{Grad } N_j \, dV, \quad G_{mi} = \int \chi_m \text{Div } N_i \, dV, \quad M_{mn} = \int \chi_m \chi_n \, dV, \quad (23)$$

with N_i the displacement basis functions, χ_m the pressure basis functions, and K the H^1 seminorm, the norm on the displacement in (18); $\ker \mathbf{G}$ is the discretely isochoric subspace, the set of displacements whose divergence has zero projection onto Q_h . The discrete inf-sup constant is $\beta_h = \sqrt{\lambda_{\min}}$ over the nonzero spectrum of

$$\mathbf{G} \mathbf{K}^{-1} \mathbf{G}^\top q = \lambda \mathbf{M} q. \quad (24)$$

Boundary data fixed everywhere always leaves one zero mode, the hydrostatic one, because $\int \text{Div } \mathbf{v} = 0$ for $\mathbf{v} \in H_0^1$; any zero beyond that is a spurious pressure mode.

8.2 Spurious modes on one element

For linear response the count of zero-energy modes is a rank count on a single element. Condensing the three fields gives

$$\mathbf{K}_e = \mathbf{K}^{\text{dev}} + \kappa \mathbf{G}^\top \mathbf{M}^{-1} \mathbf{G}, \quad M_{mn} = \int_e \chi_m \chi_n \, dV, \quad G_{mi} = \int_e \chi_m (\text{tr } \mathbf{B})_i \, dV, \quad (25)$$

with $\mathbf{B}$ the strain–displacement operator of the element and $m = \dim Q_h$ on one element. $\mathbf{K}^{\text{dev}}$ annihilates every mode with zero deviatoric strain at quadrature: the six rigid-body modes, plus the volumetric directions the displacement space can produce. The second term has rank equal to the number of those directions the pressure space resists, at most m . Anything left over carries neither deviatoric nor volumetric energy and is exactly zero-energy:

$$n_{\text{spurious}} = \dim \ker \mathbf{K}^{\text{dev}} - 6 - \text{rank } \mathbf{K}^{\text{vol}}. \quad (26)$$

For the quadratic tetrahedron $\dim \ker \mathbf{K}^{\text{dev}} = 10$: six rigid modes and four volumetric directions, the latter because $\text{tr } \boldsymbol{\epsilon}(\boldsymbol{\varphi})$ is a P_1 field. Table 4 reports (26) against the measured spectrum.

Pressure space	m	$\dim \ker \mathbf{K}^{\text{dev}}$	$\text{rank } \mathbf{K}^{\text{vol}}$	predicted	observed
P_0 constant	1	10	1	3	3
P_1 discontinuous	4	10	4	0	0
P_2 discontinuous	10	10	4	0	0

Table 4: Spurious zero-energy modes of the three-field quadratic tetrahedron at $\kappa/\mu = 10^4$, on the affine element. Script: `prototype/softmode.jl`.

Three consequences follow.

The constant-pressure element carries three modes that are exactly zero-energy on one element, not merely soft. Whether they persist after assembly is the question of §8.4.

A P_1 -discontinuous pressure removes them without an added energy term and leaves the six rigid-body modes. That is an element-level fact and no more: §8.4 and §8.5 show that the same pair locks and is unstable once assembled.

Enriching the pressure further changes nothing: the rank saturates at four, because four is all the displacement space can produce. Enriching the pressure beyond P_1^{disc} adds unknowns and leaves the rank unchanged.

The count is a property of the affine element. On a curved element, with the midside nodes displaced so that $\det \mathbf{J}$ ranges over $[0.55, 1.19]$, the isoparametric displacement space is not polynomial in $\mathbf{x}$, only the dilatation of its four volumetric directions remains free of deviatoric strain, $\dim \ker \mathbf{K}^{\text{dev}} = 7$, and the three constant-pressure modes are soft rather than exactly zero: the softest nonzero mode of the constant-pressure element lies one to two orders of magnitude below the softest mode of the linear-pressure element on the same curved element. A four-point rule samples a curved element at four points and reports the affine counts there as well; the count uses a 27-point rule. The assembled behavior of §8.4 rests on soft rather than exact modes.

Remark 7. This settles the mechanism and nothing more. Inf-sup stability is a global property of a mesh sequence and is invisible on one element; the rank argument above says which modes exist, not whether the constant in the stability estimate persists under refinement; §8.5 measures it.

8.3 A necessary condition from degree-of-freedom counts

Whether the constant in the stability estimate persists under refinement is a property of a mesh sequence, and a necessary condition for it follows from counting.

Write n_u for the number of free displacement degrees of freedom and n_p for the pressure degrees of freedom. The Schur complement of the mixed system is $\mathbf{S} = \mathbf{G}\mathbf{K}^{-1}\mathbf{G}^\top$, an $n_p \times n_p$ matrix of rank at most n_u . If $n_p > n_u$ then $\mathbf{S}$ is singular by counting, the pressure space carries modes no displacement can resist, and no inf-sup constant exists — regardless of the values those constants would otherwise take.

Table 5 counts them on a Freudenthal-subdivided cube.

Pair	N	n_u	n_p	n_p/n_u
P_2/P_0	12	36,501	10,368	0.284
P_2/P_1^{disc}	12	36,501	41,472	1.136
$P_2 \oplus b_{\text{int}}/P_1^{\text{disc}}$	12	67,605	41,472	0.613
$P_2 \oplus b_{\text{int}} \oplus b_{\text{face}}/P_1^{\text{disc}}$	12	127,221	41,472	0.326

Table 5: Degree-of-freedom counts on the unit cube, displacement fixed on the boundary. Script: `prototype/inf-sup.jl`.

For P_2/P_1^{disc} the ratio exceeds one, and its limit is computable. The quadratic nodes of this mesh family fill the $(2N+1)^3$ grid, so with the whole boundary fixed $n_u = 3(2N-1)^3 \rightarrow 24N^3$, while $n_p = 4 \cdot 6N^3 = 24N^3$ regardless of boundary data. The ratio tends to one: the pair is critically determined in the limit, which excludes a uniform inf-sup constant.

The finite- N ratio depends on the boundary condition: with the whole boundary fixed, $n_u = 3(2N-1)^3$ and the ratio exceeds one at every N ; with one face fixed, $n_u = 6N(2N+1)^2$ and it is below one at every N . Both tend to one.

Proposition 1. *On a tetrahedral mesh family with at least six tetrahedra per vertex asymptotically — the Freudenthal family among them — and the boundary fixed, P_2/P_1^{disc} is not inf-sup stable, and no refinement rescues it.*

The restriction to the mesh family is necessary. On a general tetrahedral mesh with n_V vertices, n_E edges and n_T tetrahedra, Euler’s relation for a simply connected three-dimensional mesh gives $n_E \approx n_V + n_T$, so $n_u \approx 3(2n_V + n_T)$, $n_p = 4n_T$, and $n_p/n_u \rightarrow 4/(3(2n_V/n_T + 1))$, which exceeds one only when $n_T/n_V > 6$. Below that the count refutes nothing, and on other mesh families the pair can be stable: Zhang [18] shows P_2/P_1^{disc} stable on Powell–Sabin tetrahedral meshes once their known spurious pressure modes are filtered. The general three-dimensional result is Boffi et al. [3, Remark 8.6.2]: with a P_k displacement, a discontinuous pressure is stable on general meshes only at degree $k - 3$, so for $k = 2$ nothing is admissible without enrichment.

Adding one interior bubble per element supplies $3n_{\text{el}}$ further displacement degrees of freedom and moves the ratio to 0.613; adding the four face bubbles as well — the enrichment §3.1 calls for — supplies $3n_f$ more, one vector unknown per interior face, and moves it to 0.326. Both satisfy $n_p < n_u$. The enrichment also fixes the number of global unknowns. The interior bubble condenses; the face bubbles are shared and do not. The global displacement count of the full enrichment is therefore $36,501 + 59,616 = 96,117$, which is 2.6 times the 36,501 of P_2/P_0 — also the nodal count of the composite tetrahedron on the same TETRA10 mesh — and 2.5 times the 38,698 global unknowns of Taylor–Hood, whose 2,197 vertex pressures cannot be eliminated. The 127,221 in Table 5 is the dimension of the space, not the number of global unknowns.

The count has precedents. n_p/n_u is the constraint count of Nagtegaal et al. [11], whose subject — locking of simplicial elements in the fully plastic range — is this note’s, and its reciprocal $\varrho = n_u/n_p$ is the constraint ratio of Hughes [7, §4.3.7], with $\varrho = 3$ the optimum in three dimensions. At $N = 12$: $\varrho = 3.52$ for P_2/P_0 , 0.88 for P_2/P_1^{disc} , 1.63 with the interior bubble, and 3.07 for the full enrichment. The stable pair has the optimal value, and the isochoric fraction $1 - n_p/n_u$ of §8.4 is $1 - 1/\varrho$, with $2/3$ the optimum.

Whether that limiting degeneracy is the decisive objection on the meshes tested is a separate question, which §8.4 answers.

8.4 Assembled behavior

The element count of §8.2 is a local statement. An element-local mode persists after assembly only if its trace is compatible with every neighbor across every shared face, so the global count can be smaller than $3n_{\text{el}}$, and it is what the argument against a constant pressure rests on.

Table 6 assembles $\mathbf{K}^{\text{dev}} + \kappa \mathbf{G}^T \mathbf{M}^{-1} \mathbf{G}$ over meshes under two boundary conditions: the whole boundary fixed, and only $z = 0$ fixed. The second is closer to the conditions of a body confined on part of its boundary and free elsewhere, under which soft modes are reported.

The locking measure is again a rank. A displacement mode has volumetric energy exactly when it lies outside $\ker \mathbf{G}$, so as $\kappa \rightarrow \infty$ the reachable deformations collapse onto that kernel and $\dim \ker \mathbf{G} = n_u - \text{rank } \mathbf{G}$ is the discretely isochoric subspace. A large fraction means the element still deforms; a collapse toward zero is volumetric locking.

The fraction is read against a bound rather than against intuition. Since $\text{rank } \mathbf{G} \leq n_p$, it cannot fall below $1 - n_p/n_u$. A pair that attains that bound has $\text{rank } \mathbf{G} = n_p$: every pressure mode is resisted by a distinct displacement direction. A fraction above the bound means that $n_p - \text{rank } \mathbf{G}$ pressure modes are unresisted; a pair that cannot reach it, because $n_p > n_u$, is

over-determined. The difference $n_p - \text{rank } \mathbf{G}$ counts the pressure modes no displacement resists, which for an all-Dirichlet boundary should be exactly one, the hydrostatic mode; Table 7 carries it as a column.

BC	Pair	N	n_u	n_p	$\text{rank } \mathbf{G}$	fraction	bound
all	P_2/P_0	3	375	162	158	0.579	0.568
		4	1029	384	380	0.631	0.627
	P_2/P_1^{disc}	3	375	648	363	0.032	—
		4	1029	1536	957	0.070	—
	$\oplus b_{\text{int}}$	3	861	648	644	0.252	0.247
		4	2181	1536	1532	0.298	0.296
	$\oplus b_{\text{face}}$	3	1185	648	646	0.455	0.453
		4	3045	1536	1534	0.496	0.496
	$\oplus b_{\text{int}}, b_{\text{face}}$	3	1671	648	647	0.613	0.612
		4	4197	1536	1535	0.634	0.634
	$P_2 \oplus b_{\text{face}}/P_0$	3	1185	162	161	0.864	0.863
		4	3045	384	383	0.874	0.874
$z = 0$	P_2/P_0	3	882	162	162	0.816	0.816
		4	1944	384	384	0.802	0.802
	P_2/P_1^{disc}	3	882	648	522	0.408	0.265
		4	1944	1536	1216	0.374	0.210
	$\oplus b_{\text{int}}$	3	1368	648	648	0.526	0.526
		4	3096	1536	1536	0.504	0.504
	$\oplus b_{\text{face}}$	3	1962	648	648	0.670	0.670
		4	4440	1536	1536	0.654	0.654
	$\oplus b_{\text{int}}, b_{\text{face}}$	3	2448	648	648	0.735	0.735
		4	5592	1536	1536	0.725	0.725
	$P_2 \oplus b_{\text{face}}/P_0$	3	1962	162	162	0.917	0.917
		4	4440	384	384	0.914	0.914

Table 6: Assembled count at $\kappa/\mu = 10^4$. The bound is $1 - n_p/n_u$ and is undefined where $n_p > n_u$. Assembled zero-energy modes were zero in every row. Script: `prototype/locking.jl`.

Three findings follow.

The constant-pressure soft modes do not persist after assembly. No spurious mode is present under either boundary condition at any mesh. Nor are they present as low-energy modes: the smallest eigenvalue relative to the deviatoric scale is 2.39×10^{-3} for P_0 against 2.74×10^{-3} for P_1^{disc} on the same mesh with the base alone fixed, and both are the long-wavelength modes of that block. The three modes per element of §8.2 exist on each element and are element-local: their traces do not match across faces.

The exact-zero soft modes of a constant pressure therefore do not persist after assembly on these elements. What this test does not detect is the mode Foulk III et al. [5] report, which is remedied there by a penalty and therefore has energy, small and not zero. Those are the displacement modes in $\ker \mathbf{G}_{P_0}$ whose divergence is nonzero pointwise, which carry shear energy

alone in the element; whether they are spurious is the coercivity question of §5, which §8.7 measures, and not a count of zeros. The composite tetrahedron of Foulk III et al. [5] is built on a subdivided-linear displacement space rather than P_2 , which is a different kernel and is not modeled here; the soft mode of that element may be specific to it. Nothing above contradicts their observation; it does not reproduce it on a quadratic tetrahedron.

The unenriched pair locks. With the boundary fixed, P_2/P_1^{disc} retains 3–7% of its deformation modes as κ grows, against 58–63% for P_0 ; with the base alone fixed, 37–41% against 80%. Under either boundary condition it is the pair that over-constrains, and locking is the more serious failure because the element returns a stiff response and no spurious mode indicates the error.

The interior bubble raises the rank of $\mathbf{G}$ over a linear pressure. The interior bubble vanishes on the element boundary, so its columns of $\mathbf{G}$ against a constant pressure are zero (Remark 8) and it adds nothing to the rank over P_0 . Against the linear pressure modes its columns are not zero: it raises rank $\mathbf{G}$ from 957 to 1532 at $N = 4$, and the interior-only pair has an isochoric fraction of 0.252–0.298. The full enrichment reaches 0.61–0.63, in line with P_0 and without P_0 ’s element-level degeneracy; it attains the bound $1 - n_p/n_u$ to three digits and the interior-only pair to two, so no pressure unknown is spent twice; and for the full pair $n_p - \text{rank } \mathbf{G} = 1$ at every mesh, the hydrostatic mode alone, where P_2/P_0 and the interior-only pair carry four: the enrichment raises rank $\mathbf{G}$ to $n_p - 1$.

8.5 Stability, measured against a validated control

The count of §8.3 is a necessary condition; Table 7 shows that it is not sufficient.

Two properties of the computation allow meshes to $N = 8$. With $\mathbf{K} = \mathbf{P}^\top \mathbf{L} \mathbf{L}^\top \mathbf{P}$ from a sparse Cholesky, $q^\top \mathbf{G} \mathbf{K}^{-1} \mathbf{G}^\top q = \|\mathbf{L}^{-1}(\mathbf{G}^\top q)_{[p]}\|^2$, so β_h is the smallest nonzero singular value of $\mathbf{W} = \mathbf{L}^{-1} \mathbf{G}^\top \mathbf{M}^{-1/2}$ — never forming $\mathbf{G} \mathbf{K}^{-1} \mathbf{G}^\top$, and so never squaring the condition number of the quantity measured. And the null dimension is fixed independently as $n_p - \text{rank } \mathbf{G}$, so the cut is taken by index rather than by selecting a threshold. The gap is unambiguous: $O(10^{-15})$ below it against $O(10^{-2})$ above.

The sweep carries a positive control, the Taylor–Hood pair, whose stability in three dimensions is established [3]: a valid test must report it stable. A negative control alone cannot detect a test that reports every pair unstable; the negative control, P_1/P_0 , gives $\beta_h = 0.333, 0.229, 0.165, 0.129$ at $N = 2$ to 5 with 45, 138, 303, 558 unresisted pressure modes, the alternating-sign spurious pressure mode of that pair (script `prototype/infsup.jl`).

The control is reported stable: β_h rises and settles on 0.2216, flat to three digits over the last three refinements, which validates the reading of the remaining rows.

Every row of Table 7 is what Boffi et al. [3] predict. What the sweep establishes is that the scripts are sound and that the theory applies to this mesh family; what it adds to the theory is the constants. The rows are ordered as the construction of Boffi et al. [3, Example 8.7.2] adds degrees of freedom.

P_2 does not control a constant pressure in three dimensions. P_2/P_0 decays at a local rate approaching one, although $n_p/n_u = 0.284$, and carries four pressure null modes at every mesh where one is allowed: the hydrostatic mode and three exact spurious modes that no

Pair	$h = 1/2$	$1/4$	$1/6$	$1/8$	null	local rate
P_2/P_1 continuous (control)	0.1734	0.2186	0.2214	0.2216	1	−0.003
P_2/P_0	0.1001	0.0755	0.0544	0.0418	4	0.97
$\oplus b_{\text{int}}$	0.1015	0.0756	0.0544	—	4	0.84
$\oplus b_{\text{face}}$	0.4671	0.4228	0.4060	0.3967	1	0.076
P_2/P_1^{disc}	0.1132	0.0424	0.0277	—	$n_p - n_u$	1.04
$\oplus b_{\text{int}}$	0.0782	0.0564	0.0406	—	4	0.83
$\oplus b_{\text{face}}$	0.0938	0.0701	0.0497	—	2	0.88
$\oplus b_{\text{int}}, b_{\text{face}}$	0.2875	0.2962	0.2968	—	1	−0.003

Table 7: β_h over a Freudenthal mesh sequence, all-Dirichlet, ordered as the construction of Boffi et al. [3, Example 8.7.2] runs. “null” is $n_p - \text{rank } \mathbf{G}$, the pressure modes no displacement resists, which must be one; “local rate” is $\ln(\eta_N/\eta_{N+1})/\ln((N+1)/N)$ at the last refinement, constant for a genuine power law and shrinking toward zero for a sequence approaching a limit. Script: `prototype/beta.jl`.

quadratic displacement resists. That is Boffi et al. [3, §8.7] verbatim — “to control piecewise constant pressure, we need some degrees of freedom on the faces” — and the three modes are the missing face control made visible.

The interior bubble cannot supply it. $P_2 \oplus b_{\text{int}}/P_0$ reproduces P_2/P_0 to three digits with the same four null modes, for the reason in Remark 8. Over P_1^{disc} the interior bubble changes β_h from 0.0424 to 0.0564 at $h = 1/4$ and from 0.0277 to 0.0406 at $h = 1/6$, and the rate falls from 1.04 to 0.83, but the sequence decays and four null modes remain. This pair has $n_p/n_u = 0.613$ (Table 5) and is unstable.

Face bubbles control the constant pressure. $P_2 \oplus b_{\text{face}}/P_0$ drops the null dimension to one at every mesh and its β_h approaches from above with a local rate that falls from 0.16 to 0.08 over the sequence, below the constant rate of a power law and above the rate −0.003 of the control. Five to seven points cannot separate a limit near 0.35 from a slow decay on their own, and the row is reported as consistent with stable; its stability rests on Boffi et al. [3, Example 8.7.2], whose Fortin construction is exactly what the face degrees of freedom make possible. Over P_1^{disc} the face bubbles alone leave one spurious linear pressure mode per mesh and β_h decays: controlling the linear modes is the function of the interior bubble.

Both together are the three-dimensional Crouzeix–Raviart pair, and it is stable. β_h rises monotonically to 0.2968, flat to three digits over the last three refinements, with a local rate of −0.003 — the same rate as the control’s, and at a higher constant than the control’s 0.2216. The pressure null space is one dimensional at every mesh. Each part of the enrichment controls one part of the pressure space, and neither controls the other’s.

Remark 8 (What separates the two enrichments). The interior bubble has zero pairing with a constant pressure: it vanishes on the whole element boundary, so $\int_{\Omega_e} \text{Div } \mathbf{b}_{\text{int}} = 0$ and its columns of $\mathbf{G}$ against a P_0 mode are identically zero. It pairs only with the linear pressure

modes, which is why it changes P_2/P_1^{disc} and changes the P_2/P_0 constant by at most 0.0014 (the $P_2 \oplus b_{\text{int}}/P_0$ row of Table 7, with the same null dimension): upward, because enlarging U_h cannot lower β_h , and by that little, because the bubble's columns of $\mathbf{G}$ against P_0 are zero. `checks.jl` reads those columns directly and confirms it. The face bubbles are what supply a displacement response to the pressure jump across a face, which is the mode a discontinuous pressure needs resisted.

P_2/P_0 and $P_2 \oplus b_{\text{int}}/P_1^{\text{disc}}$ have $n_p/n_u = 0.284$ and 0.613 and are unstable: a ratio above one excludes a pair, and a ratio below one does not admit it.

Remark 9 (Mesh dependence). These results are for the Freudenthal subdivision, a structured family, and structured meshes are where pair counterexamples characteristically arise: the alternating-sign spurious pressure mode of P_1/P_0 has its largest amplitude on them. A pair can degenerate on a structured mesh and remain well behaved on a general one. On this family, P_2/P_0 , P_2/P_1^{disc} and $P_2 \oplus b_{\text{int}}/P_1^{\text{disc}}$ decay; confirming that as a property of the pairs rather than of the mesh needs a sequence of unstructured meshes, which this study does not have. §8.6 gives one measurement in the opposite sense: on the non-affine images of this mesh the P_2/P_0 constant is smaller. The affirmative result needs no such confirmation, since Boffi et al. [3, Example 8.7.2] establish it on any shape-regular family; what the sweep adds is the constant.

8.6 Stability on a deformed configuration

Every measurement above is at $\mathbf{F} = \mathbf{I}$. At a finite deformation the pairing is $\int \varpi \operatorname{div}_{\mathbf{x}} \psi \, dV$ (§3.1), the divergence pairing on the deformed configuration with the pressure scaled by $1/J$, and whether β_h stays bounded below under a large non-affine deformation is what the regime of §1 requires. The operators (23) are assembled on the isoparametric image of the quadratic mesh under a prescribed map, with the derivatives, the divergence and the volume taken on the image and the pressure basis on the reference cell as the element uses it, and with the boundary conditions of the reference cube; the scaling by $1/J$ changes β_h by at most the ratio of extreme Jacobians and is not applied. Four maps of the unit cube, about its center: a twist about the vertical axis by 90° and by 180° over the height, isochoric and non-affine, with shear strains near 2 at the corners of the latter; a parabolic shear $x_1 = X_1 + \frac{1}{2}X_3^2$, isochoric; a radial inflation with J from 1 to 2.3; and an affine compression to $J = \frac{1}{2}$ as a control, since an affine map changes β_h only through the element aspect ratio. Table 8 gives the result at $N = 5$ with the local rate over the last refinement. Script: `prototype/deformed.jl`.

Three findings.

β_h is not uniform in the deformation. It decreases with the local shear of the map for every pair, and most for the Crouzeix–Raviart pair: from 0.297 to 0.204 under the 90° twist, 0.116 under the 180° twist and 0.183 under the parabolic shear, against $0.221 \rightarrow 0.189, 0.137, 0.176$ for Taylor–Hood and $0.413 \rightarrow 0.385, 0.345, 0.336$ for the face-bubbled constant-pressure pair. The inflation, which changes J by a factor of 2.3 but shears little, leaves the constant at 0.276. The affine compression gives 0.213, flat to four digits from $N = 3$: the aspect-ratio effect alone. The dependence is on the shear of the deformation and not on its volume change.

At a fixed deformation the sequence in h is consistent with a positive limit. The local rate of the Crouzeix–Raviart pair falls with N on every non-affine map: 0.28, 0.22, 0.18

Table 8: β_h at $N = 5$ on deformed configurations, whole boundary fixed, with the local rate over $N = 4 \rightarrow 5$ in parentheses for the Crouzeix–Raviart pair; the null dimension is one on every row. The P_2/P_0 column gives the pressure null dimension and β_h : its three exact spurious modes exist only on the affine mesh.

map	J	P_2/P_1 cont.	$P_2 \oplus b_{\text{face}}/P_0$	$P_2 \oplus b_{\text{int}} \oplus b_{\text{face}}/P_1^{\text{disc}}$	P_2/P_0 (null, β_h)
reference	1	0.221	0.413	0.297 (−0.003)	4, 0.063
twist 90°	0.99–1.01	0.189	0.385	0.204 (0.18)	1, 0.027
twist 180°	0.98–1.05	0.137	0.345	0.116 (0.42)	1, 0.051
parabolic shear	1	0.176	0.336	0.183 (0.05)	1, 0.005
inflation	1.0–2.3	0.192	0.368	0.276 (0.01)	1, 0.004
compression	0.5	0.192	0.333	0.213 (0.00)	4, 0.038

over $N = 2$ to 5 for the 90° twist, 0.67, 0.53, 0.42 for the 180° twist, and 0.05 for the parabolic shear (through $N = 5$). A power-law decay would hold its rate. The 180° row has not flattened by $N = 5$, and its shear strain of about 2 at the corners is beyond what a plastic zone of the benchmark set reaches; the finding is that the constant degrades with the strain, not that it vanishes. At $N = 6$ the pair gives 0.1975, 0.1088 and 0.182 on the three non-affine maps, local rates of 0.17, 0.35 and 0.04, each below the rate before it.

The exact spurious modes of P_2/P_0 belong to the affine mesh. Under every non-affine map the pressure null dimension of P_2/P_0 falls from four to one, the three modes are resisted, and β_h of P_2/P_0 is smaller on every non-affine image, 0.004 to 0.05, than on the reference mesh, 0.063.

8.7 Coercivity under a plastic tangent

The claim of §5 is measured as the constant it names. With $\mathbf{Z}$ a basis of $\ker \mathbf{G}_h$,

$$\lambda_{\min} = \min_q \frac{q^\top \mathbf{Z}^\top \mathbf{K}^{\text{ep}} \mathbf{Z} q}{q^\top \mathbf{Z}^\top \mathbf{K}^{H^1} \mathbf{Z} q}, \quad \mathbf{K}_{ij}^{\text{ep}} = \frac{1}{2\mu} \int \boldsymbol{\epsilon}_i : \mathbb{C}^{\text{ep}} : \boldsymbol{\epsilon}_j \, dV, \quad (27)$$

is the smallest generalized eigenvalue of the deviatoric form under the tangent (21) against the H^1 seminorm, restricted to the discretely isochoric subspace. It is dimensionless, it is taken on the exact kernel rather than through a penalty, and it is the constant of the second hypothesis; (22) says what to expect of it, at least $(1 - \eta)/2$ for every pair when $\eta < 1$ and a decay to zero for every pair when $\eta = 1$. The quantity that carries the claim is the dilatation of the eigenmodes,

$$r_v(\mathbf{v}) := \frac{\int (\text{Div } \mathbf{v})^2 \, dV}{\int |\text{Grad } \mathbf{v}|^2 \, dV}, \quad (28)$$

zero for an isochoric field. The flow direction is uniform, $\mathbf{n} = \text{sym}(\mathbf{e}_1 \otimes \mathbf{e}_2)/\sqrt{2}$ or $\mathbf{n} = \text{dev}(\mathbf{e}_3 \otimes \mathbf{e}_3)/\|\cdot\|$, which is the most unfavorable choice: the continuum operator then admits shear bands of any width, a decaying eigenvalue is expected of every pair, and only the dilatation of the mode says whether the band is physical. The tangent is taken at $\eta = 1$ and at $\eta = 0.9$, the boundary is fixed everywhere, and each pair is measured elastically ($\eta = 0$) on the same kernel. Script: `prototype/plastic.jl`.

Table 9: Lowest mode of the deviatoric form on $\ker \mathbf{G}_h$ under the tangent (21) at $\eta = 1$, all-Dirichlet Freudenthal cube: the eigenvalue λ_1 of (27) and the dilatation r_v of its mode, (28), for a shear and a uniaxial flow direction. The elastic lowest mode has $\lambda_1 = 0.500$ and $r_v = 0$ on every row, as the identity behind (22) requires. At $\eta = 0.9$ every λ_1 exceeds the floor 0.05 and the dilatations agree with those below to two digits. Script: `prototype/plastic.jl`.

Pair	N	$\mathbf{n} = \text{sym}(\mathbf{e}_1 \otimes \mathbf{e}_2)/\sqrt{2}$		$\mathbf{n} = \text{dev}(\mathbf{e}_3 \otimes \mathbf{e}_3)/\ \cdot\ $	
		λ_1	r_v	λ_1	r_v
P_2/P_0	3	0.109	0.065	0.117	0.208
	4	0.067	0.034	0.106	0.217
	5	0.047	0.029	0.101	0.220
$P_2 \oplus b_{\text{face}}/P_0$	3	0.044	0.044	0.082	0.274
	4	0.026	0.028	0.074	0.268
	5	0.017	0.019	0.071	0.261
$P_2 \oplus b_{\text{int}} \oplus b_{\text{face}}/P_1^{\text{disc}}$	3	0.074	0.036	0.147	0.018
	4	0.049	0.019	0.139	0.0074
	5	0.032	0.0066	0.134	0.0042
P_2/P_1 continuous (control)	3	0.056	0.051	0.053	0.861
	4	0.030	0.029	0.026	0.929
	5	0.019	0.020	0.016	0.956

The scripts are checked by an identity. By the identity behind (22), the elastic form on any kernel satisfies $\lambda = \frac{1}{2} + \frac{1}{6}r_v$, so its lowest mode has $r_v = 0$ and $\lambda_1 = 0.5$ exactly. Every row of the sweep reproduces this; at $N = 2$ for P_2/P_0 , where the kernel has no isochoric member, $\lambda_1 = 0.5024$ with $r_v = 0.0146$, as the identity requires. The identity is also the check on the energy metric of the scripts.

The constant does not separate pairs, as (22) predicts. Under the shear direction λ_1 decays for every pair, the control included, at rates between $h^{1.5}$ and h^2 : the discrete spaces resolve narrower shear bands, which the continuum operator admits with zero energy. Under the uniaxial direction no shear band has $\epsilon \propto \mathbf{n}$, and λ_1 converges for the three element-local pairs, to 0.10 for P_2/P_0 , 0.071 for $P_2 \oplus b_{\text{face}}/P_0$ and 0.13 for the Crouzeix–Raviart pair, while it decays for Taylor–Hood. At $\eta = 0.9$ every λ_1 lies above the floor of 0.05, the smallest being the control’s 0.067.

The dilatation of the soft modes separates them, in the direction claimed. Under the uniaxial direction the lowest modes of both constant-pressure pairs carry $r_v = 0.20$ – 0.27 at every refinement, with no decay. These are the modes of §5: the element assigns them no energy for a dilatation the continuum resists with κ , and they are the softest modes the element has. The lowest modes of the Crouzeix–Raviart pair carry 0.040, 0.018, 0.0074 and 0.0042 over $N = 2$ to 5, decaying as h^2 , the rate at which Boffi et al. [3, Prop. 5.1.3] makes $\ker \mathbf{G}_h$ approximate the isochoric space; its soft modes are physical. Under the shear direction the lowest modes of every pair are shear bands whose dilatation decays, the Crouzeix–Raviart pair’s three to four times smaller than the others’ at $N = 5$. Counted over the twenty lowest

Table 10: Spurious modes among the twenty lowest of the plastic form on $\ker \mathbf{G}_h$ ($r_v > 0.1$) in a confined plastic zone with a varying flow direction, at $N = 3, 4, 5$.

field	boundary	P_2/P_0	$P_2 \oplus b_{\text{face}}/P_0$	$P_2 \oplus b_{\text{int}} \oplus b_{\text{face}}/P_1^{\text{disc}}$	P_2/P_1 cont.
indentation	all fixed	16, 15, 7	18, 17, 5	4, 0, 0	20, 18, 17
indentation	base fixed	11, 1, 0	13, 5, 0	0, 0, 0	14, 7, 5
twist	all fixed	14, 18, 15	18, 13, 8	1, 4, 0	20, 20, 20
twist	base fixed	5, 2, 0	13, 7, 4	0, 0, 0	12, 13, 12

modes, with a mode called spurious when $r_v > 0.1$ (the stable pair’s soft modes lie below 0.02 from $N = 4$ on, the spurious ones at 0.2–0.4): under the uniaxial direction at $N = 5$ all twenty of each constant-pressure pair’s and of Taylor–Hood’s are spurious, and none of the Crouzeix–Raviart pair’s at any N ; under the shear direction the twenty lowest of every element-local pair are bands at $N = 5$. The spurious modes of a constant pressure are therefore present in both directions but are the lowest modes only where the flow admits no band; the next paragraph therefore counts them among the twenty lowest in a zone with a varying flow direction.

A confined zone with a varying flow direction. The flow direction and the plastic zone are taken from a prescribed displacement field: $\mathbf{n} = \text{dev } \boldsymbol{\epsilon}(\mathbf{u}) / \|\text{dev } \boldsymbol{\epsilon}(\mathbf{u})\|$ at each quadrature point and $\eta = 1$ where $|\text{dev } \boldsymbol{\epsilon}(\mathbf{u})|$ exceeds 0.35 of its peak, $\eta = 0$ elsewhere. Two fields: an indentation of the top face, $u_3 = -\exp(-d^2/w^2) X_3$ with d the distance from the vertical axis and $w = \frac{1}{4}$, whose zone is 51% of the volume with elastic surroundings; and the displacement of the 90° twist of §8.6, whose zone is 97%. Both are run with the whole boundary fixed and with the base alone fixed. Table 10 counts the spurious modes among the twenty lowest. Script: `prototype/plastic_zone.jl`.

With the whole boundary fixed the constant-pressure pairs retain between five and fifteen spurious modes among their twenty softest at $N = 5$, and under the twist the softest mode of P_2/P_0 is spurious at every N up to 4 ($r_v = 0.22$ – 0.31) and its third at $N = 5$ (0.20). The Crouzeix–Raviart pair has none from $N = 4$ on under the indentation and none at $N = 5$ under the twist, and none at any N with the base alone fixed. Taylor–Hood keeps seventeen to twenty, and under the twist its softest mode’s dilatation grows with refinement, 0.29, 0.37, 0.66, 0.74. With the base alone fixed the softest modes of every pair are long-wavelength motions of the block and the counts fall for the constant-pressure pairs too, to zero at $N = 5$ for P_2/P_0 and four for the face-bubbled pair, so the spurious modes are then above the twentieth; they are not absent, as the all-fixed rows show. What claim 4 of §9 adds is whether a nonlinear solve excites them.

Taylor–Hood is the worst pair on this measure. Its pressure is continuous, so its kernel is divergence-free only against continuous linear test functions, and under the uniaxial direction its lowest modes are dilatation almost entirely: $r_v = 0.86, 0.93, 0.96$ at $N = 3, 4, 5$, with λ_1 decaying to 0.016. Those modes have no counterpart in the continuum, which resists them with κ . Taylor–Hood is therefore an alternative for elasticity only (§10).

Table 11: Quadrature rules on the enriched element: zero-energy modes of the condensed stiffness (six is correct) and of the deviatoric stiffness (ten: six rigid modes and the four conformal fields), and the extreme softening factors against a 216-point rule, on the reference tetrahedron and on a Kuhn tetrahedron of the Freudenthal mesh. $\kappa/\mu = 10^4$. Script: `prototype/quadrature.jl`.

rule	points	reference			Kuhn		
		zero($\mathbf{K}_e$)	min	max	zero($\mathbf{K}_e$)	min	max
RFE degree 2	4	21	0	1.30	21	0	1.30
RFE degree 3	5	21	< 0	1.13	21	< 0	1.12
conical degree 3	8	6	0.10	1.14	6	0.03	1.21
Keast degree 5	14	6	1.00	1.35	6	1.00	1.35
conical degree 5	27	6	0.91	1.00	6	0.88	1.00
conical degree 7	64	6	1.00	1.00	6	1.00	1.00

8.8 Quadrature rule and basis

Two implementation choices set the number of operations per element and its explicit stable step, and neither is decided by the stability results: the quadrature rule, and the basis that carries the enriched space. Both are measured on one element. Scripts: `prototype/quadrature.jl` and `prototype/basis.jl`.

Quadrature. The scripts integrate the enriched element to degree 6 so that the linear measurements are exact. For the element the criterion is weaker: a rule is admissible when the condensed stiffness $\mathbf{K}_e = \mathbf{K}^{\text{dev}} + \kappa \mathbf{G}^T \mathbf{M}^{-1} \mathbf{G}$ has exactly six zero-energy modes, and its effect is the softening it introduces, measured as the extreme generalized eigenvalues of $\mathbf{K}_e(\text{rule})$ against $\mathbf{K}_e(\text{exact})$ on the complement of the rigid modes, which are 1 for an exact rule. Table 11 gives both for the rules considered, with a degree-11 rule of 216 points as the reference. The two rules supplied by `ReferenceFiniteElements`, the Julia package that provides the reference elements and their quadrature, four and five points, leave 21 zero modes and are inadmissible. The eight-point conical rule (a tensor-product Gauss rule mapped onto the tetrahedron) is rank sufficient and softens the element to 0.03–0.10 of its stiffness in the worst direction, which is also inadmissible. Two degree-5 rules are admissible: the 14-point symmetric rule of Keast [8], which integrates every monomial through degree 5 to 4×10^{-16} and over-stiffens the interior bubble’s own block by 35% since its gradient-squared is of degree 6, and the 27-point conical rule, which under-stiffens the same block by 9–12%. Assembled, both give the Crouzeix–Raviart pair’s constant within 1% (0.293 and 0.298 against 0.297 at $N = 5$), a one-dimensional pressure null space, no spurious mode among the twenty lowest under the uniaxial plastic tangent, and the same dilatation of the lowest mode (0.0073, 0.0074 and 0.0074). The 14-point rule satisfies the criteria — six zero-energy modes, positive weights, 4.6 times fewer points than the exact rule — and is used, with a stiffer interior bubble.

Basis. The space $P_2 \oplus b_{\text{int}} \oplus b_{\text{face}}$ can be carried by the hierarchical basis of the scripts, ten Lagrange functions and five bubbles, or by the nodal basis with unit value at one of fifteen nodes, the `TETRA15` arrangement: vertices, edge midpoints, face centroids, centroid. The space is the same, so every stiffness and stability result is the same; what differs is conditioning,

Table 12: Basis choice on the reference tetrahedron, $\kappa/\mu = 2$, $\mu = \rho = 1$: condition numbers of the condensed stiffness on the complement of the rigid modes and of the consistent mass, the smallest lumped mass over ρV , the momentum of a unit translation under the lumped mass over ρV , and the Courant number $c_p \Delta t/h$ from the element bound. Script: `prototype/basis.jl`.

basis, lumping	cond $\mathbf{K}_e$	cond $\mathbf{M}$	min $m/\rho V$	momentum	Courant
hierarchical, consistent	797	778	—	—	0.092
hierarchical, HRZ over all 15	—	—	0.014	0.486	0.097
hierarchical, HRZ over Lagrange 10	—	—	0.028	1.000	0.114
nodal, consistent	559	94	—	—	0.092
nodal, HRZ	—	—	0.021	1.000	0.145
TETRA10, HRZ	—	—	0.028	1.000	0.197
TETRA10, row sum	—	—	−0.050	1.000	—
TETRA4, row sum	—	—	0.250	1.000	0.476

lumping, and the coefficients of a rigid translation. Table 12 gives the measurements at $\kappa/\mu = 2$. The nodal basis conditions the consistent mass eight times better (94 against 778) and the stiffness better (559 against 797). Under HRZ lumping the hierarchical basis loses 51% of the momentum of a rigid translation unless the normalization runs over the Lagrange functions alone (§6); the nodal basis conserves it without qualification, and its lumped masses are positive on every element measured. The explicit stable step from the element bound, as the Courant number $c_p \Delta t/h$, is 0.476 for TETRA4, 0.197 for TETRA10 under HRZ, 0.145 for the enriched element in the nodal basis under HRZ, and 0.114 in the hierarchical basis under Lagrange-only HRZ, with the same ordering on the distorted and Kuhn elements. The nodal basis is used: a node-based degree-of-freedom manager carries it without a new kind of unknown, its mass lumps correctly by the standard construction, and it takes a 20–27% larger stable step.

9 Summary of the evidence and open measurements

The formulation rests on five claims. For each, the section that establishes it and the number that carries it are given; the measurements that remain are listed at the end.

The pair is inf-sup stable in three dimensions. §8.5: the Crouzeix–Raviart pair has $\beta_h = 0.2968$ at a local rate of -0.003 , above the Taylor–Hood control’s 0.2216, with a one-dimensional pressure null space at every mesh, and every other element-local pair decays. The face bubbles are the necessary part of the enrichment; the interior bubble alone is not sufficient (Remark 8). Uniformity in κ/μ follows from stability of the incompressible limit [3, §5.5.2, §8.12.1, Remark 8.12.1]. On deformed configurations (§8.6) the constant decreases with the shear of the deformation, to 0.204 at a 90° twist and 0.116 at 180° , is unchanged by a volume change of 2.3, and at each fixed deformation the sequence in h is consistent with a positive limit; the constant is not uniform in the deformation.

No spurious soft modes. In elasticity the second hypothesis of Brezzi’s theorem holds for every conforming pair (§5). The three exact zero-energy modes per element of a constant

pressure (§8.2) do not persist after assembly (§8.4). Under a plastic tangent (§8.7) the coercivity constant is bounded below by (22) for every pair, and the dilatation of the soft modes separates them: 0.20–0.27 without decay for the constant-pressure pairs, h^2 decay for the Crouzeix–Raviart pair, and in a confined zone with a varying flow direction no spurious mode among the twenty softest of the Crouzeix–Raviart pair at $N = 5$ against five to fifteen for the constant-pressure pairs.

No volumetric locking. §8.4: the discretely isochoric fraction $\dim \ker \mathbf{G}/n_u$ is 0.61–0.63 for the Crouzeix–Raviart pair, at the bound $1 - n_p/n_u$, against 0.03–0.07 for P_2/P_1^{disc} , which locks.

Element-local elimination. §3.2: the pressure is discontinuous, so its elimination is element by element and the tangent has the sparsity of the displacement mesh; the face bubbles add global displacement unknowns, 2.6 times those of P_2/P_0 at $N = 12$ (§8.3).

Quadrature and basis. §8.8: the 14-point degree-5 rule keeps six zero-energy modes and the assembled constant within 1%; the nodal basis conserves the momentum of a rigid translation under HRZ lumping and gives a Courant number of 0.145 against 0.476 for TETRA4.

Open measurements. Three remain. An unstructured mesh sequence for the negative stability results, which are established here on the Freudenthal family only (Remark 9). The smoothness of the equivalent plastic strain in a nonlinear solve, on the benchmark set on which the composite tetrahedron performs poorly [5]: punch indentation, a nearly incompressible Cook’s membrane in three dimensions (a tapered panel clamped on one edge and sheared on the opposite one), large-rotation twist of a bar, and a weak shock in a material of high κ/μ ; this is to be measured with the Simo–Hughes J_2 model implemented in Norma and Carina rather than with the additive logarithmic plasticity of §4 (Remark 6), against the composite tetrahedron on the same meshes. And the sparsity of the assembled tangent in an implementation, which §3.2 establishes by construction.

10 Limitations

This section states what the formulation requires, what it does not establish, and what its parts are.

The number of unknowns. The face bubbles of the Crouzeix–Raviart pair are shared between neighbors and do not condense: at $N = 12$ the enriched space carries 96,117 global displacement degrees of freedom against 36,501 for P_2/P_0 and for the composite tetrahedron on the same mesh, a factor of 2.6, and 2.5 times Taylor–Hood’s 38,698 (§8.3). Whether the accuracy of a stable pair on coarser meshes compensates for that factor is not established; for elasticity that is the comparison with Taylor–Hood, whose pressure is global and whose displacement space is 2.5 times smaller, and for plasticity Taylor–Hood’s softest modes are dilatation almost entirely (§8.7), so the comparison there is with the composite tetrahedron.

Operations per element. With the 14-point rule of §8.8 the enriched element evaluates the constitutive law at 14 points on 45 displacement degrees of freedom, against 4 or 5 points on 30 for TETRA10 and one on 12 for TETRA4. The number of elements a quadratic tetrahedron needs for a given accuracy in this problem class is not measured.

The explicit stable step. HRZ lumping in the nodal basis gives positive masses and conserves momentum, and the Courant number is 0.145 against 0.476 for TETRA4 on the same element, a factor of 3.3, of which 2.4 is the quadratic element and 1.4 the enrichment (§8.8). The comparison at equal accuracy, on the coarser mesh a quadratic element allows, is not measured; if weak shocks are run with the implicit integrator the factor does not apply.

The material. §2.5 requires an exact volumetric–isochoric split. Of the seven materials implemented in Norma, five satisfy it; linear elasticity and Saint-Venant–Kirchhoff do not, and are not finite-deformation models in the sense of §1. The three with a volumetric energy that is not quadratic require one projection more (Remark 1). Anisotropic or non-separable response is outside the split and would need the five-field variant of §7 or a different treatment. The $\bar{F}$ family of elements carries no such requirement.

The parts. Mixed volumetric treatment [16], logarithmic-strain plasticity [9] and displacement–pressure pairs satisfying (18) uniformly [3] are established. The contribution is their combination: an element-local pair satisfying (18) uniformly, a projected volumetric strain, and a deviatoric law evaluated pointwise, which removes the stabilization that the composite-tetrahedron and variational-multiscale methods carry. The evidence is that of §8; the nonlinear confirmation is the open measurement of §9. Face-hybridized methods [1] are the alternative should that measurement not confirm it.

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